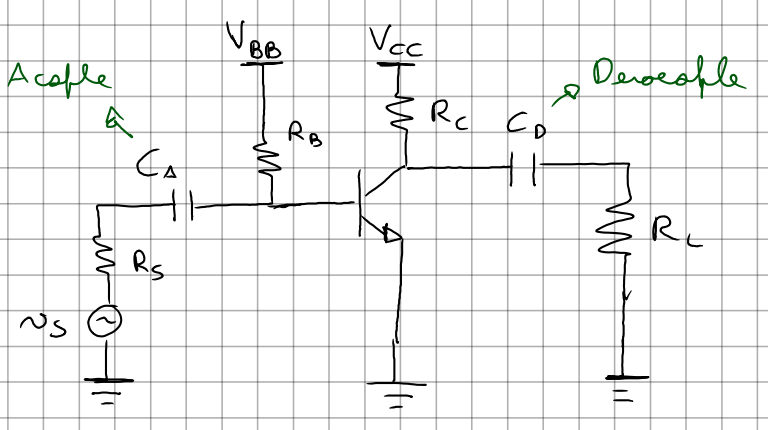
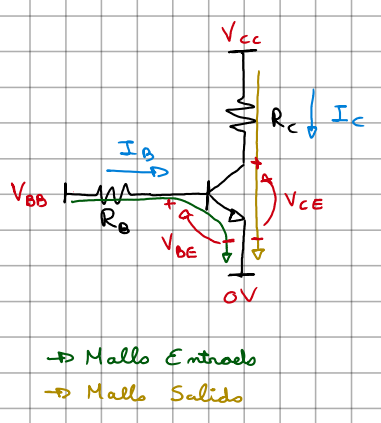
TBJ:



Analizo Polarizacion (DC):

Cap → Abiertos



→ Malla Entrada
→ Malla Salida

$M_e)\ V_{BB} - I_B \cdot R_B - V_{BE} = 0$

$M_s)\ V_{CC} - I_C \cdot R_C - V_{CE} = 0$

Hip. MAD: $V_{BEQ} = 0{,}7$; $I_{CQ} = \beta I_{BQ}$

, Q → Punto de Trabajo (I_{CQ}, V_{CEQ})

$M_e \rightarrow V_{BB} - \frac{I_{CQ}}{\beta} \cdot R_B - 0{,}7V$

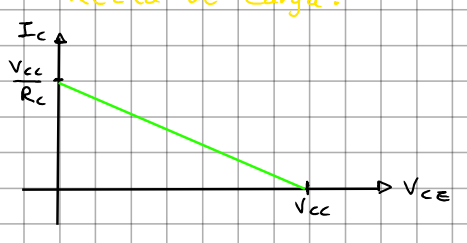
$\rightarrow I_{CQ} = \frac{V_{BB} - 0{,}7V}{R_B/\beta}$ ✓

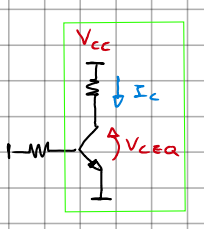
$M_s \rightarrow V_{CC} - I_{CQ} \cdot R_C - V_{CE} = 0$

$\rightarrow V_{CE} = V_{CC} - I_{CQ} \cdot R_C$ ✓

Verifico MAD → $V_{CEQ} > V_{CE(Sat)} \approx 0{,}2V$ ✓

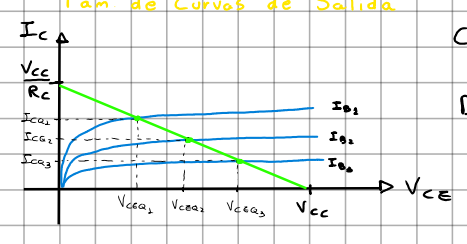
Recta de Carga:





$\rightarrow I_C = \frac{V_{CC} - V_{CE}}{R_C}$

Fam de Curvas de Salida



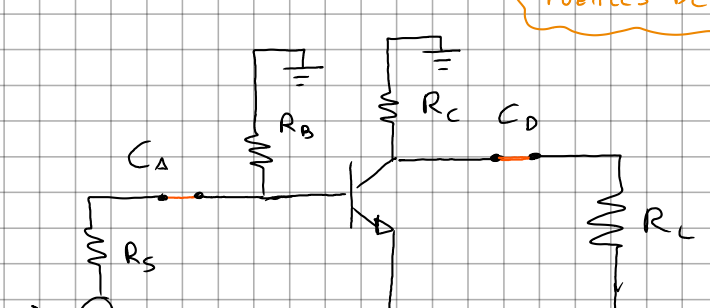
Como la $I_C = I_B \cdot \beta$

Depende de I_B en el $Q = (I_{CQ}, V_{CEQ})$ q tengo.

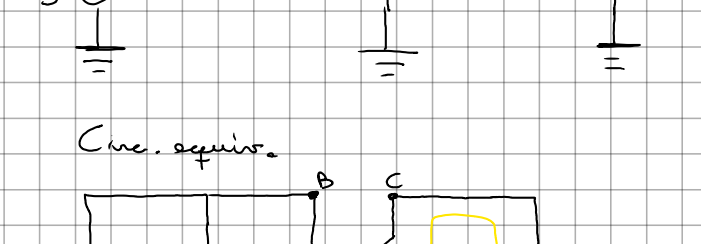
Analizo PS (AC):

Cap → Cortos

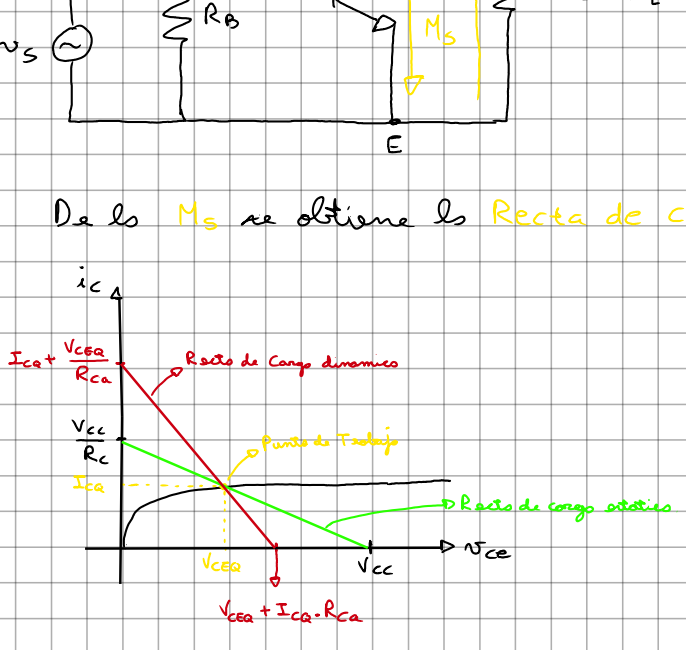
Fuentes DC → 0V



Circ. equiv.

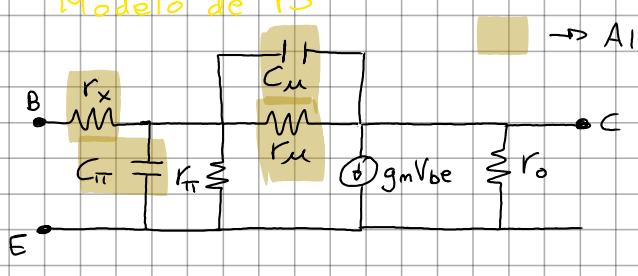


De los M_s se obtiene la Recta de carga dinamica



Modelo de PS

→ Altas frec.



Formulas:

- $g_m = \frac{I_{CQ}}{V_{Th}}$ → Transconductancia
- $r_\pi = \frac{V_{Th}}{I_{BQ}} = \beta \frac{V_{Th}}{I_{CQ}} = \frac{\beta}{g_m}$ → R. de entrada
- $r_o = \frac{V_A}{I_{C(MAD)}} \approx \frac{V_A}{I_{CQ}}$ → R. de salida
- $r_\mu = \beta_o \cdot r_o$ → R. de realimentacion → Muy grande (≈ Abierto)
- $r_x =$ (?)
- $C_\pi = C_{dBE} = \tau_T \cdot g_m$ → C. Base - Emisor
- $C_\mu = C_{jBC} = \frac{C_{jBC_o}}{\sqrt{1 + \frac{V_{CB}}{\Phi_B}}}$ → C. Base - Colector

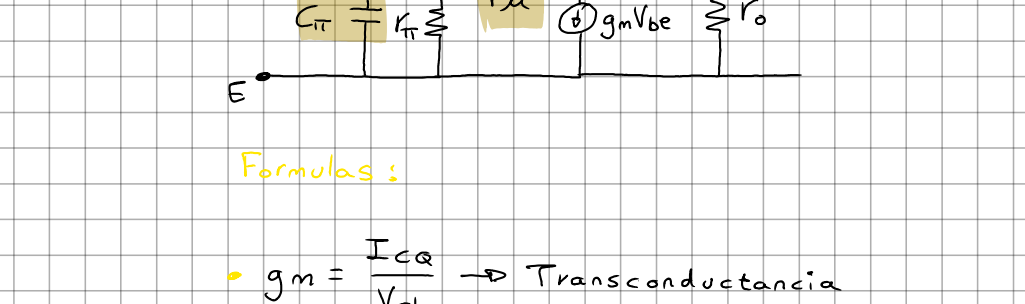
Unilateral?

Simplificaciones para Emisor Comun. Creo q mediana y baja frecuencia

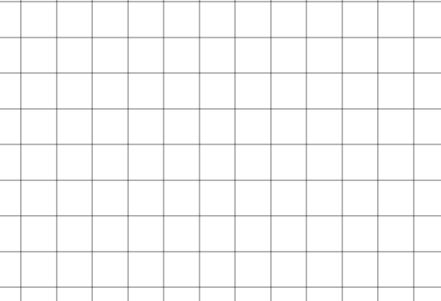
→ Despreciar efectos Reactivos, C_π y C_μ → Abiertos

→ $r_\mu \rightarrow \infty$

→ $r_x \rightarrow 0$



Modelo Amp



$$A_V = \frac{v_{out}}{v_{in}} = \frac{-g_m \cdot v_{be} \cdot R_{ca}}{v_{be}} = -g_m \cdot R_{ca}$$

Si R_S es comparable a R_{in}

$$A_{V_S} = \frac{v_o}{v_s} = \frac{v_i}{v_s} \cdot \frac{v_o}{v_i} = A_V \cdot \frac{R_{in}}{R_{in} + R_S}$$

$= A_V$

$v_i = \frac{R_{in}}{R_{in} + R_S} \cdot v_s$, $R_{in} = R_B /\!/ r_\pi$